INF10025 Data Management and Analytics

Task 2 – Pass Submission

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Student Number (VietNam/Australia): SWS01217/ 105565520

A. Task Pass2a

In the first task of the Pass section, students are required to open file named ***studentlist.csv*** by **Notepad** editor.

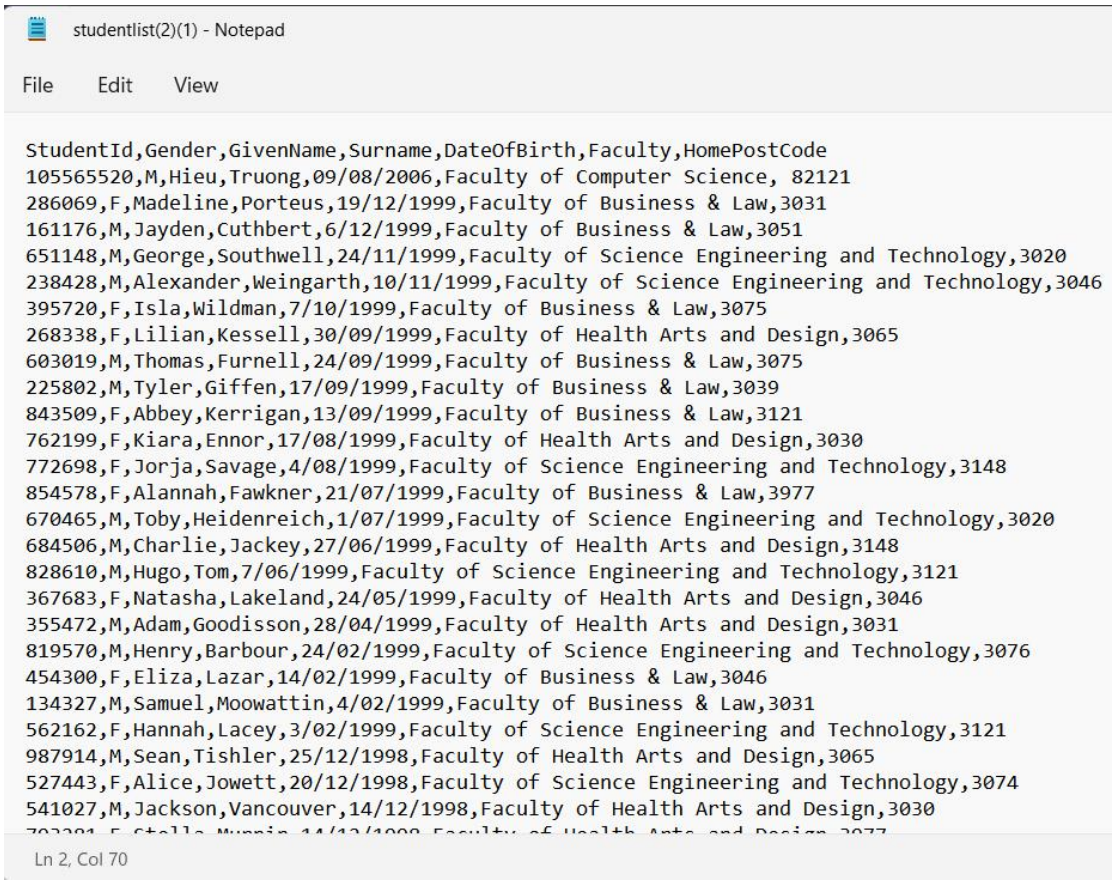


Figure 1: Editing studentlist.csv in Notepad

Next, select Delimited, choose ***Comma (Tick First Row Contains Field Names)***, and set table name is ***Student5520*** in order too import file ***studentlist.csv*** into ***.accdb*** file

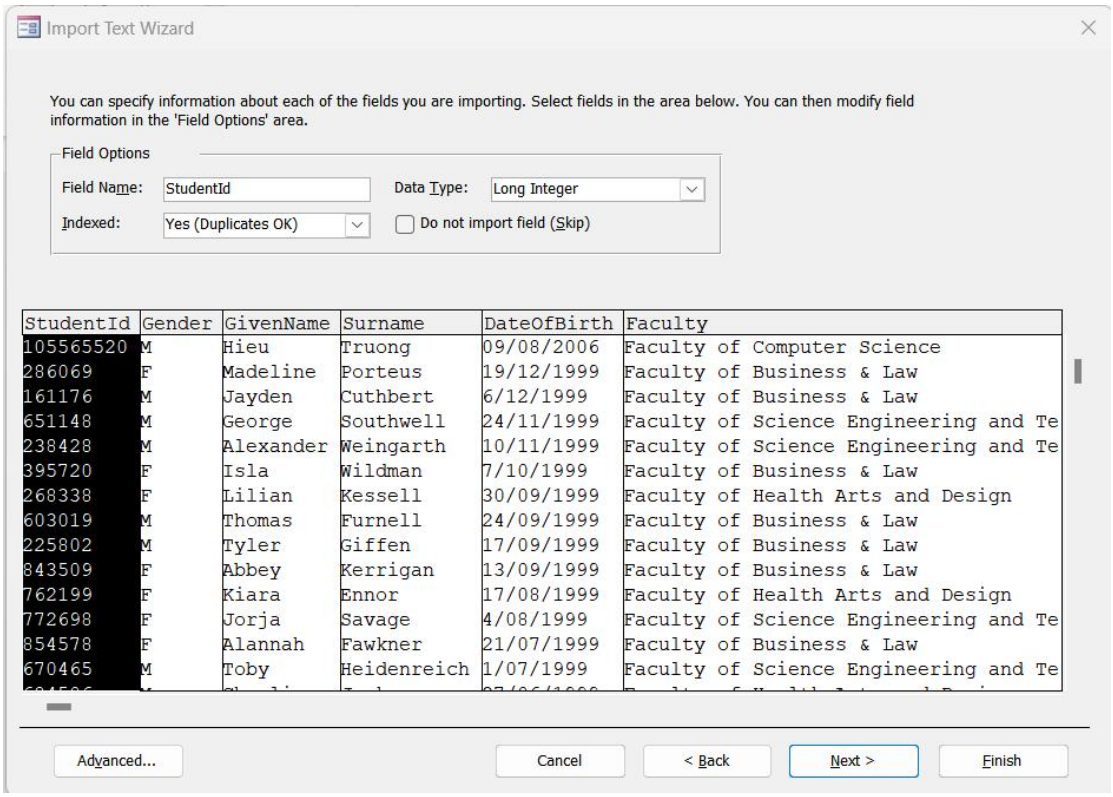


Figure 2: Field Options Process in Import Text Wizard

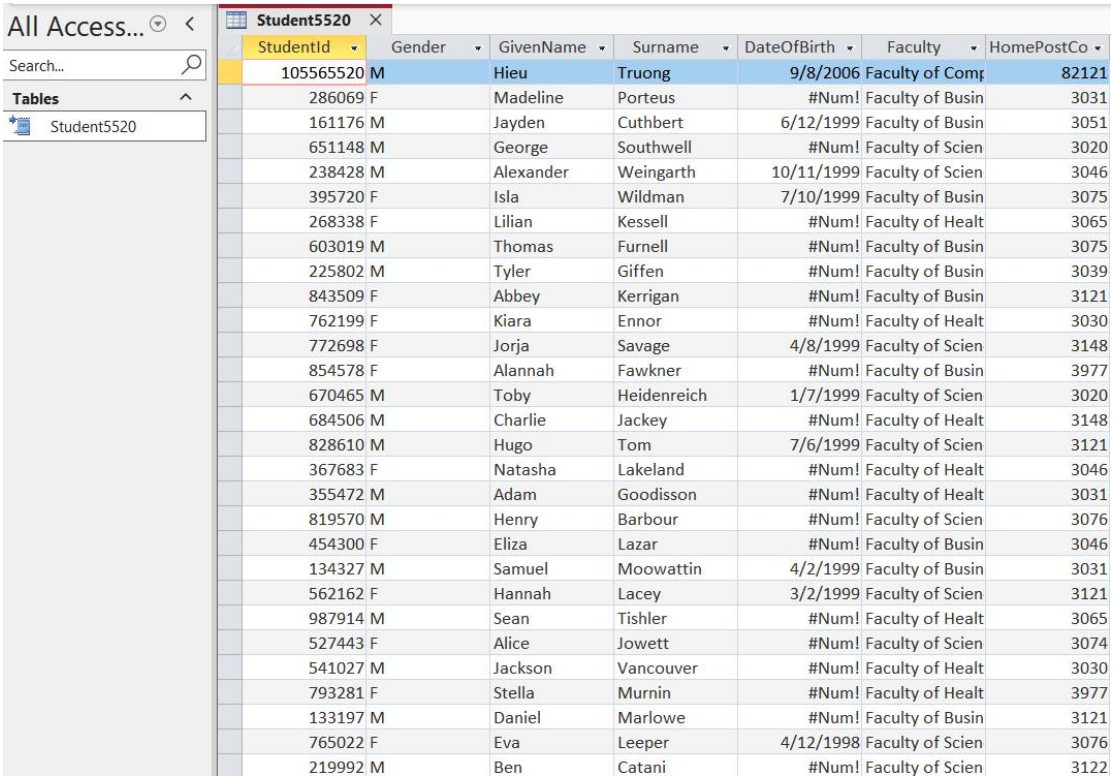


Figure 3: Datasheet View of Student5520 Table

B. Task Pass2b

- In my opinion, double-clicking typically opens a ***.csv*** file in Microsoft Excel because Windows operating system the ***.csv*** file extension with Excel by default. Furthermore, when Excel is installed, it registers itself with Windows as the default program for handling Comma-Separated Values (***.csv***) files because tabular spreadsheet software is commonly used to view and edit this type of structured data.

- For permanently change the default program for ***.csv*** files to Notepad:

1. Right click on any ***.csv*** files
2. Select Open with and then click on Choose another app (or *More apps*).
3. Scroll down and select Notepad from the list of available applications.
4. Check the box that says "Always use this app to open ***.csv*** files".
5. Click OK. From then on, double-clicking any ***.csv*** file will automatically open it in Notepad instead of Excel.

C. Task Pass2c

In the third task, students must switch to ***Design View***, add the field name ***FeesPaid***, and set its data type to ***Currency***.

Student5520

Field Name	Data Type
StudentId	Number
Gender	Short Text
GivenName	Short Text
Surname	Short Text
DateOfBirth	Date/Time
Faculty	Short Text
HomePostCode	Number
FeesPaid	Currency

Figure 4: Table Design View with FeesPaid Field

Next, open the datasheet view of the ***Student5520*** table and sort the data in descending order by ***StudentId*** (displaying the top 10 rows with the largest student numbers at the top).

Student5520

StudentId	Gender	GivenName	Surname	DateOfBirth	Faculty	HomePostCo	FeesPaid
105565520	M	Hieu	Truong	9/8/2006	Faculty of Comp	82121	
993207	M	Samuel	Kenyon	9/9/1997	Faculty of Healt	3056	
987914	M	Sean	Tishler	#Num!	Faculty of Healt	3065	
965653	F	Brianna	Salomons	#Num!	Faculty of Busin	3020	
960671	M	Max	Lawless	11/7/1996	Faculty of Healt	3040	
943722	M	Lang	Jaw-Long	#Num!	Faculty of Scien	3053	
935171	M	Max	Permewan	#Num!	Faculty of Busin	3064	
934378	F	Indiana	McDonagh	9/1/1999	Faculty of Healt	3032	
924210	M	Nicholas	Thorn	#Num!	Faculty of Busin	3053	
894264	M	Dominic	Muskett	#Num!	Faculty of Healt	3040	

Figure 5: Top 10 Rows of Student5520 in Descending Student ID Sequence

Subsequently, students are required to generate a copy table name ***STUCOPY5520***



StudentId	Gender	GivenName	Surname	DateOfBirth	Faculty	HomePostCo
105565520	M	Hieu	Truong	9/8/2006	Faculty of Comp	82121
286069	F	Madeline	Porteus	#Num!	Faculty of Busin	3031
161176	M	Jayden	Cuthbert	6/12/1999	Faculty of Busin	3051
651148	M	George	Southwell	#Num!	Faculty of Scien	3020
238428	M	Alexander	Weingarth	10/11/1999	Faculty of Scien	3046
395720	F	Isla	Wildman	7/10/1999	Faculty of Busin	3075
268338	F	Lilian	Kessell	#Num!	Faculty of Healt	3065
603019	M	Thomas	Furnell	#Num!	Faculty of Busin	3075
225802	M	Tyler	Giffen	#Num!	Faculty of Busin	3039
843509	F	Abbey	Kerrigan	#Num!	Faculty of Busin	3121
762199	F	Kiara	Ennor	#Num!	Faculty of Healt	3030
772698	F	Jorja	Savage	4/8/1999	Faculty of Scien	3148
854578	F	Alannah	Fawkner	#Num!	Faculty of Busin	3977
670465	M	Toby	Heidenreich	1/7/1999	Faculty of Scien	3020

Figure 6: Navigation Pane Showing Both Student5520 and STUCOPY5520 Tables.

D. Task Pass2d

Before generating the query, the requirement instructs students to close ***Student5520*** table and then create query named ***T2PD_Update_5520.*** Then, in the ***Query Design*** tab, click on the ***Update*** to specify the update rule, and set based on following instructions:

- ***Field***: FeesPaid, ***Table***: Student5520, ***Update To***: 4000.
- ***Field***: Gender, ***Table***: Student5520, ***Criteria***: “M”.

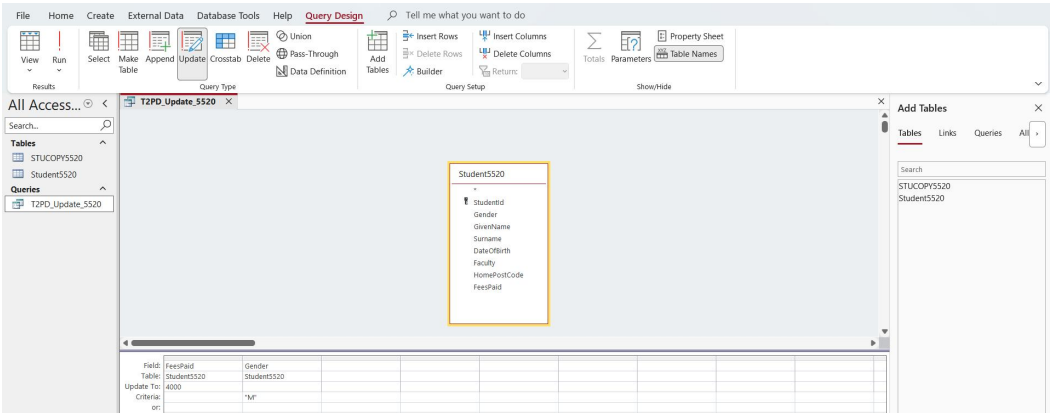


Figure 7: Query Design (Query Configuration) for T2PD_Update_5520

After successful configuration, reopen the ***Student5520*** table, sort the ***StudentId*** in ***descending*** order, and verify that the updated ***FeesPaid*** values are correctly displayed for all male students.

Student5520

StudentId	Gender	GivenName	Surname	DateOfBirth	Faculty	HomePostCo	FeesPaid	Click to Add
105565520	M	Hieu	Truong	9/8/2006	Faculty of Computer Science	82121	$4,000.00	
993207	M	Samuel	Kenyon	9/9/1997	Faculty of Health Arts and Design	3056	$4,000.00	
987914	M	Sean	Tishler		Faculty of Health Arts and Design	3065	$4,000.00	
965653	F	Brianna	Salomons		Faculty of Business & Law	3020		
960671	M	Max	Lawless	11/7/1996	Faculty of Health Arts and Design	3040	$4,000.00	
943722	M	Lang	Jaw-Long		Faculty of Science Engineering and Technology	3053	$4,000.00	
935171	M	Max	Permewan		Faculty of Business & Law	3064	$4,000.00	
934378	F	Indiana	McDonagh	9/1/1999	Faculty of Health Arts and Design	3032		
924210	M	Nicholas	Thorn		Faculty of Business & Law	3053	$4,000.00	
894264	M	Dominic	Muskett		Faculty of Health Arts and Design	3040	$4,000.00	

Figure 8: Updated Top 10 Rows of Student5520 in Descending Student ID Sequence

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E. Task Pass2e

Subsequently, the requirement guide student to close the student table first, choose ***Query Design*** in the ***Create*** ribbon, select ***Delete*** in the ***Query Type,*** and configure these following:

- ***Field***: Surname,
- ***Table***: Student5520.
- ***Delete***: Where.
- ***Criteria***: Like “M*”

After configuring above requirements, save query named ***T2PE_Delete_5520***.

Field:	Surname
Table:	Student5520
Delete:	Where
Criteria:	Like "M*"
or:	

Figure 9: Query Design for T2PE_Delete_5520

After setting configure successfully, click on ***Run*** button. After that, choose table ***Student5520*** table on Datasheet View and sort StudentID Descending.

Student5520

StudentId	Gender	GivenName	Surname	DateOfBirth	Faculty	HomePostCo	FeesPaid	Click to Add
105565520	M	Hieu	Truong	9/8/2006	Faculty of Computer Science	82121	$4,000.00	
993207	M	Samuel	Kenyon	9/9/1997	Faculty of Health Arts and Design	3056	$4,000.00	
987914	M	Sean	Tishler		Faculty of Health Arts and Design	3065	$4,000.00	
965653	F	Brianna	Salomons		Faculty of Business & Law	3020		
960671	M	Max	Lawless	11/7/1996	Faculty of Health Arts and Design	3040	$4,000.00	
943722	M	Lang	Jaw-Long		Faculty of Science Engineering and Technology	3053	$4,000.00	
935171	M	Max	Permewan		Faculty of Business & Law	3064	$4,000.00	
924210	M	Nicholas	Thorn		Faculty of Business & Law	3053	$4,000.00	
884570	M	George	Yirawala		Faculty of Science Engineering and Technology	3058	$4,000.00	
870470	F	Dakota	Hassell		Faculty of Science Engineering and Technology	3148		

Figure 10: Updated Top 10 Rows of Student5520 After Deleting Students with Surname Starting with 'M'

In detail, when comparing the ***Student5520*** table to the ***STUCOPY5520*** backup table, 11 records with a surname starting with '***M***' have been successfully deleted.

F. Task Pass2f

Similarly to Task Pass2e, close the student table first, create a new query design named ***T2PF_Param_5520***, and configure it according to the following instructions:

- ***Field***: StudentId, ***Table***: Student5520
- ***Field***: GivenName, ***Table***: Student5520
- ***Field***: SurName, ***Table***: Student5520
- ***Field***: Gender, ***Table***: Student5520
- ***Field***: HomePostCode, ***Table***: Student5520, ***Criteria***: [Enter Postcode:]

Field:	StudentId	GivenName	Surname	Gender	HomePostCode
Table:	Student5520	Student5520	Student5520	Student5520	Student5520
Sort:					
Show:	☑	☑	☑	☑	☑
Criteria:					[Enter Postcode:]
or:					

Figure 11: Query Design Grid for Parameter Query T2PF_Param_5520

After configuring the above settings, click the ***Run*** button to display the ***Enter Parameter Value*** dialog box that prompts the user to enter a postcode.

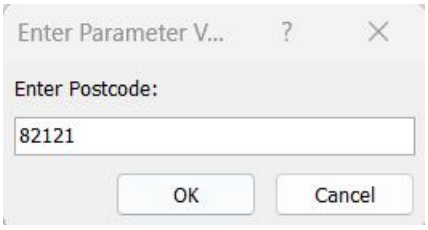
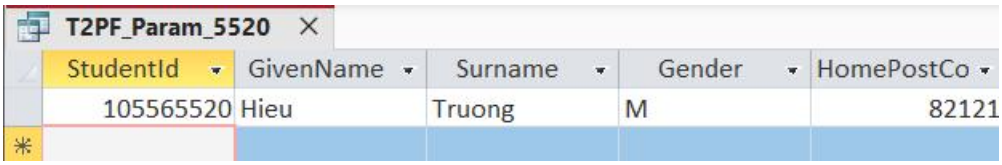


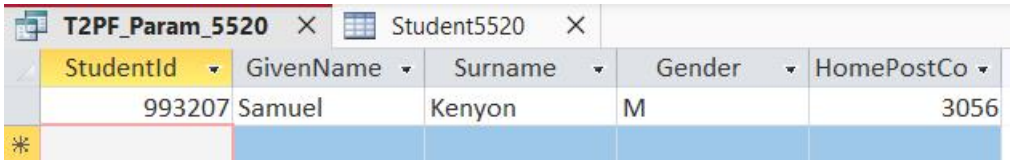
Figure 12: Enter Parameter Value Dialog Box Prompting for Postcode Input



StudentId	GivenName	Surname	Gender	HomePostCo
105565520	Hieu	Truong	M	82121

Figure 13: Query Result for T2PF_Param_5520 Using Postcode 82121

Next, testing the query with another postcode (***3056***) yields the following result:



StudentId	GivenName	Surname	Gender	HomePostCo
993207	Samuel	Kenyon	M	3056

Figure 14: Query Result for T2PF_Param_5520 Using Postcode 3056

G. Task Pass2g

In Task ***Pass2g***, the requirement instructs students to get familiar with Microsoft Power BI. First and foremost, double-click on the Microsoft Power BI icon to initiate the program, choose ***Get Data*** from other sources, select ***Database*** options, select ***Connect,*** click on ***moviedatabaseDAD,*** choose ***Open,*** and finally right-mouse click on ***Import*** all six tables from the ***moviedatabaseDAD.accdb*** file obtained in ***Task 01***.

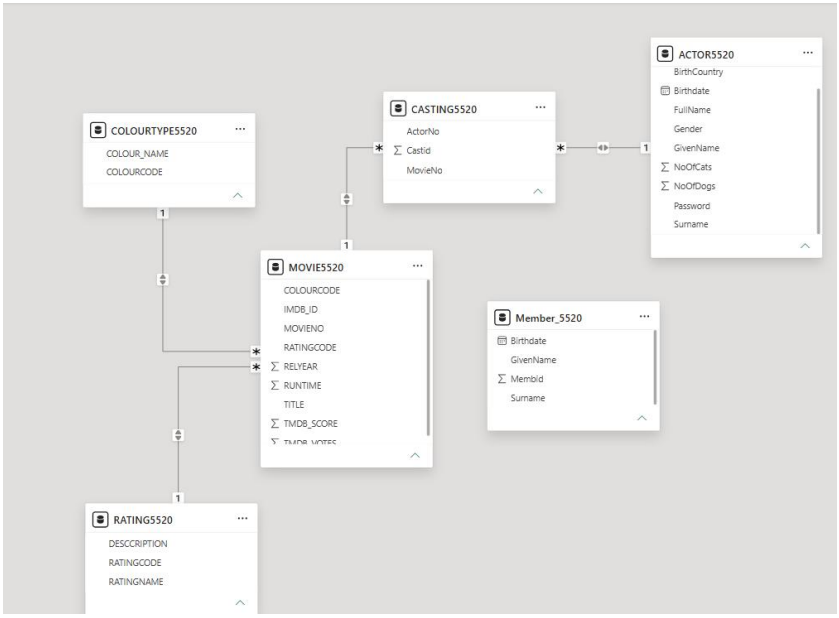


Figure 15: Power BI Model View Showing Table Relationships and Cross-Filter Settings

H. Task Pass2h

Regarding the requirement of Task Pass2h, I changed to ***Report View***, clicked the ***Table*** icon on the ***Visualizations*** pane, and added the four following fields:

TITLE, ***RELYEAR***, ***RUNTIME***, and ***RATINGCODE***. Then, under the Build visual tab below the ***Visualizations***, choose ***Don't Summarize*** for each field.

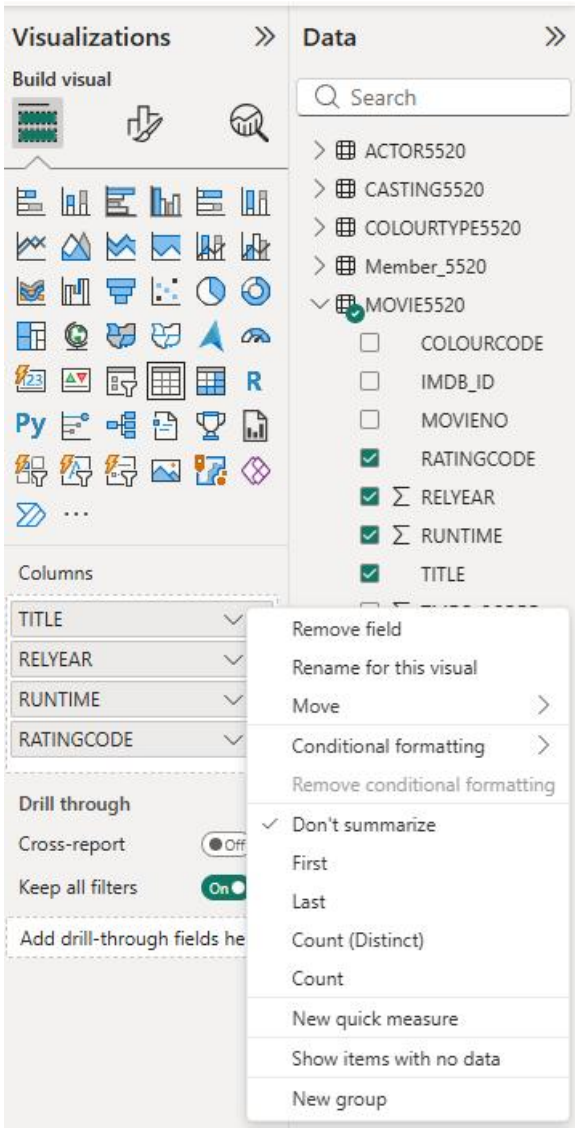


Figure 16: Configuring Table Fields

In terms of sorting ***RELYEAR*** in ascending order, I clicked on the ***RELYEAR*** column header in the table until the upward-pointing arrow appears.

TITLE	RELYEAR	RUNTIME	RATIN
Casino Royale	2006	144	M
Happy Feet	2006	108	G
Little Miss Sunshine	2006	102	PG
Night at the Museum	2006	108	PG
The Black Dahlia	2006	121	MA
The Da Vinci Code	2006	149	M
The Departed	2006	151	M
The Devil Wears Prada	2006	109	M
The Pink Panther	2006	93	PG
You, Me and Dupree	2006	108	M
Evan Almighty	2007	96	PG
Harry Potter and the Order of the Phoenix	2007	138	PG
Hot Fuzz	2007	121	MA
Juno	2007	96	PG
Live Free or Die Hard	2007	128	M
Ocean's Thirteen	2007	122	M
Spider-Man 3	2007	139	PG
The Bucket List	2007	97	M
Transformers	2007	144	PG
Wild Hogs	2007	100	M
Burn After Reading	2008	96	MA
Get Smart	2008	110	PG
Iron Man	2008	126	PG
Kung Fu Panda	2008	90	PG

Figure 17: Table visual displaying the filtered and sorted movie data.

Regarding of adding a heading to the visualization that includes my Student ID and title, I clicked ***Format your visual*** under the Visualizations pane, turned on the ***Title***, and enter Text named ***105565520 - Movie Data (Released from 2006 Onward)*** with font is ***Times New Roman***, font size: ***15***, Text Color: ***Brown***, and Horizontal Alignment is ***Center***

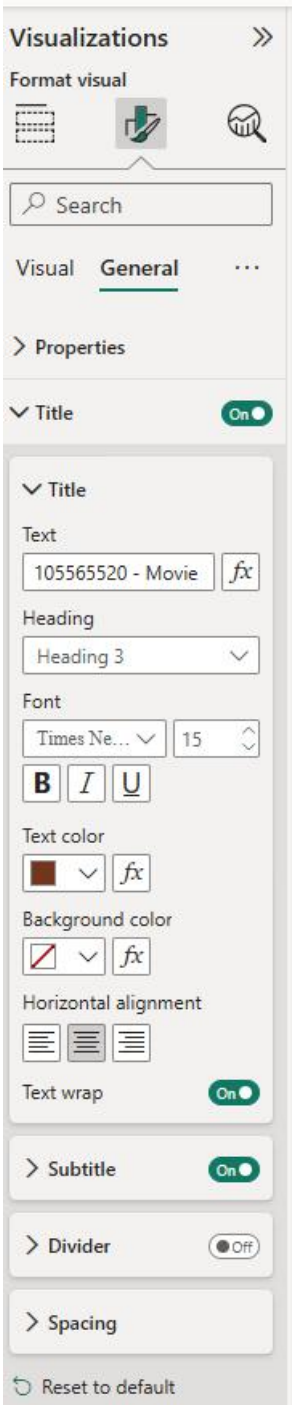


Figure 18: Configuring visual title formatting

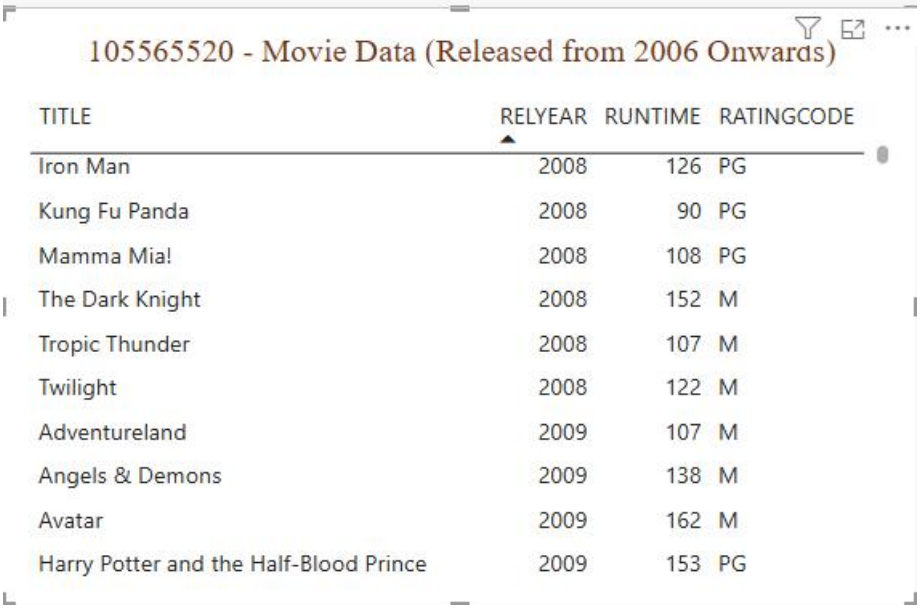


Figure 19: Filtered and sorted movie table visual with custom title

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I. Task Pass2i

In this section, task Pass2i asks students to create a Donut chart with following settings:

- The donut chart must display the count of Movies by RatingCode
- Add a heading to the visualization that includes your student ID and the title of the visualization
- Format Title Font size and colour to your liking.
- The data labels must show both the Rating Code and the Count of movies.

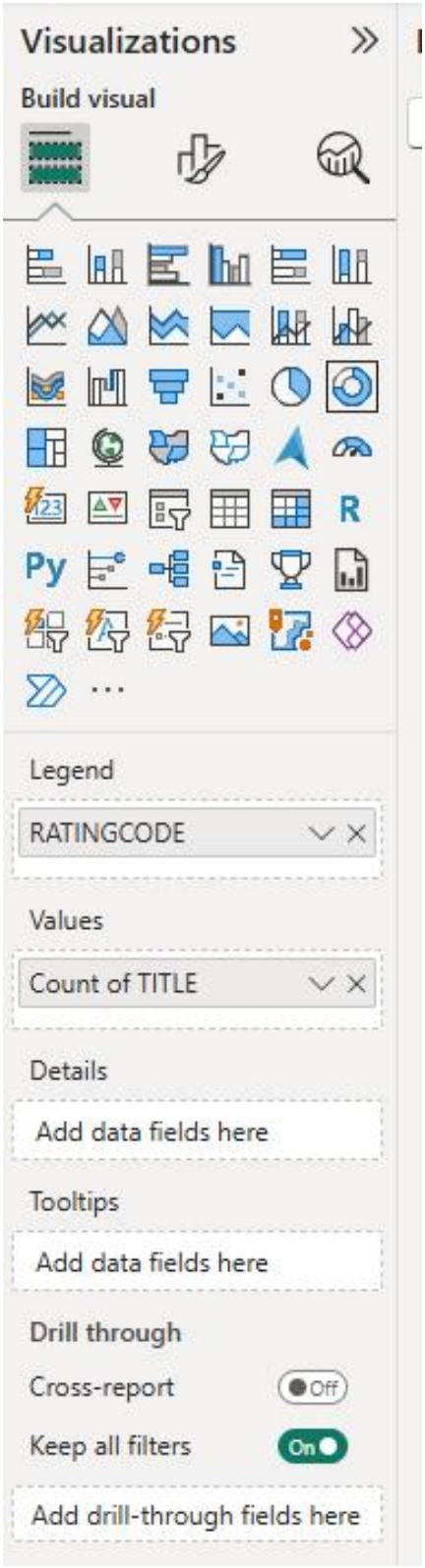


Figure 20: Configuring Donut Chart Fields and Values

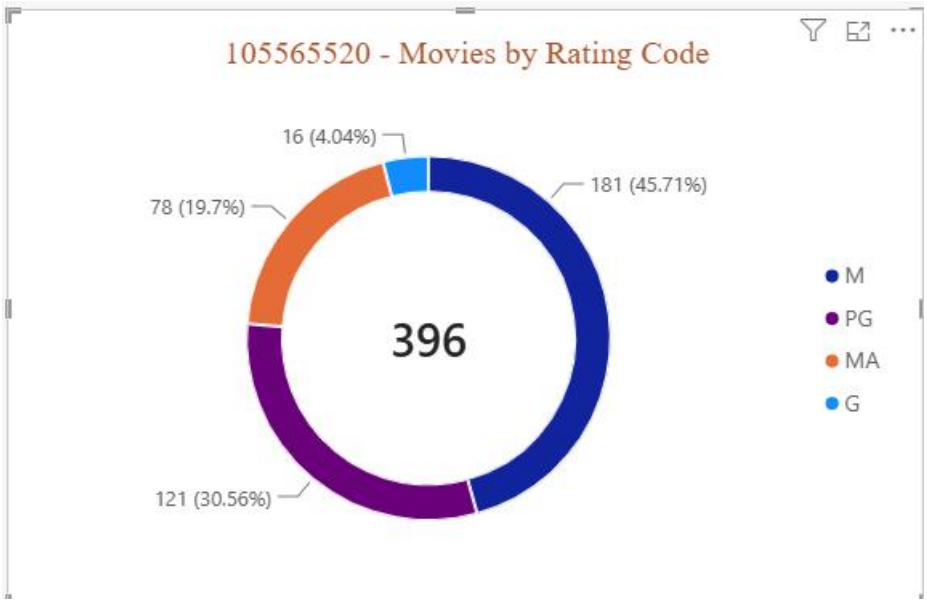


Figure 21: Donut chart displaying movies by rating code with custom title

According to ***Figure 21***, the donut chart illustrates the distribution of movies categorized by their rating codes (M, PG, MA, and G). Firstly, the M (Mature) rating accounts for the highest proportion with 181 movies (45.71%), followed by PG (Parental Guidance) with 121 movies (30.56%). While the MA (Mature Accompanied) and G (General) ratings, 78 movies (19.7%) and 16 movies (4.04%), respectively, bringing the total number of movies analyzed to 396.

J. Task Pass2j

For Task Pass2j, I put ***TITLE*** from the ***MOVIE5520*** table into ***Rows***, ***Gender*** from the ***ACTOR5520*** table into ***Columns***, and ***ActorNo*** from the ***ACTOR5520*** table into ***Values*** for generating the Matrix visualization. After that, I set the title to ***105565520 - Actor Count by Movie and Gender*** with a font size of 15, an orange color, and center alignment.

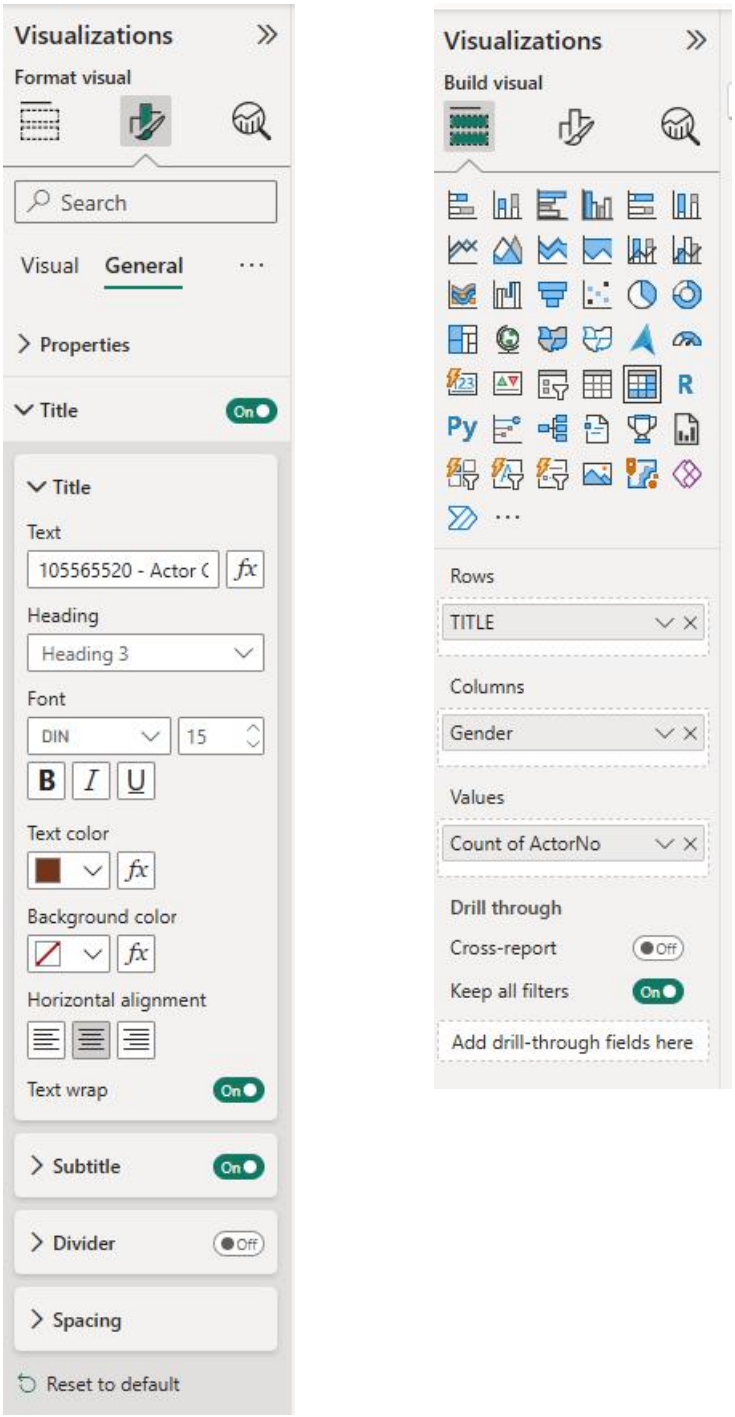


Figure 22: Visualizations Pane Settings (Build visual and Format visual tabs)

105565520 - Actor Count by Movie and Gender

TITLE	F	M	Total
50 First Dates	2	7	**9**
A Few Good Men		6	**6**
A League of Their Own	3	3	**6**
A Star Is Born		2	**2**
About a Boy	3	2	**5**
Ad Astra		3	**3**
Adventureland	2	3	**5**
Alexander and the Terrible, Horrible, No Good, Very Bad Day	3	4	**7**
American Hustle	2	5	**7**
Total	**224**	**603**	**827**

Figure 23: Matrix Visualization of Actor Count by Movie and Gender

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K. Task Pass2k

In the Pass2k part, the task requires students to generate a Map visualization with the following settings:

- It must display the count of actors by birth country.
- Add a heading to the visualization that includes your student ID and the title of the visualization

However, I had to go to ***File,*** choose ***Options and settings,*** click on ***Options,*** select ***Security,*** and tick on ***Use Map and Filled Map visuals.*** In ***Build Visual*** part, choose ***BirthCountry*** from ***ACTOR5520*** table for ***Location*** field, ***FullName*** from ***ACTOR5520*** table for ***Legend*** field. In ***General*** part of the ***Format Visual,*** I set Title is ***105565520 - Actor Count by Birth Country, Font size*** is ***15*** and ***Center Alignment.***

Step 29: ***Setting Configuration***

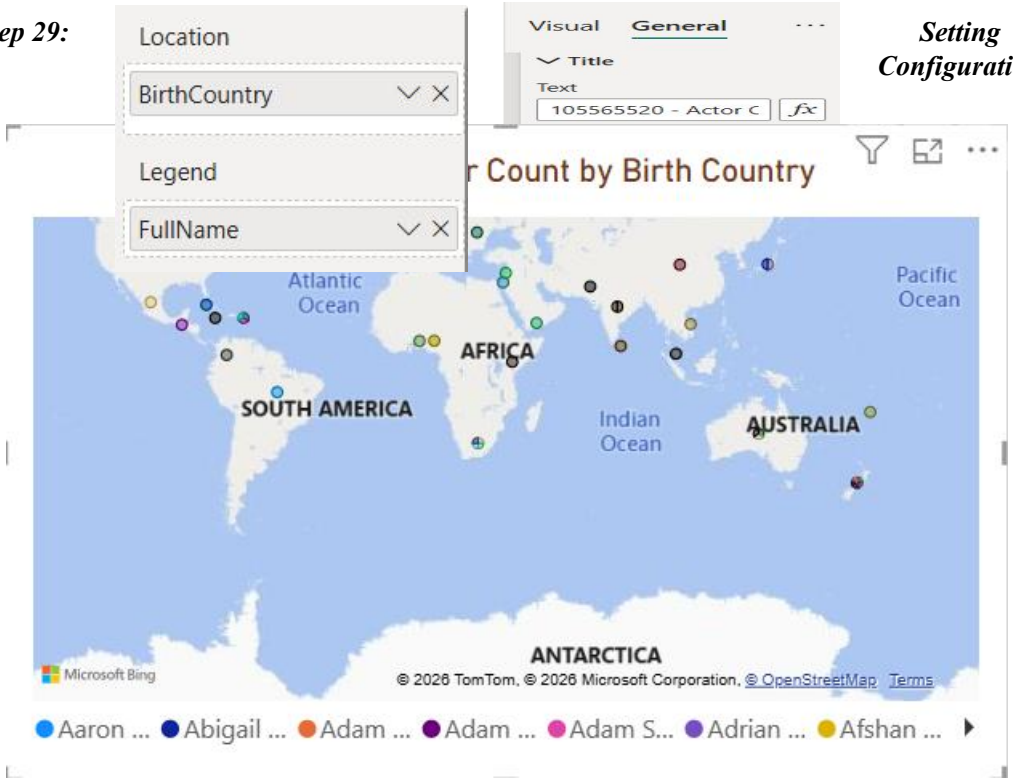


Figure 30: Map visualization showing the count of actors by birth country.

L. Task Pass2l

In this section, I generated a Clustered Bar Chart with following configuration:

- ***Y-axis***: BirthCountry
- ***X-axis***: Count of ActorNo
- ***Title***: 105565520 - Actor Count by Birth Country
- ***Font Size***: 15
- ***Alignment***: Center

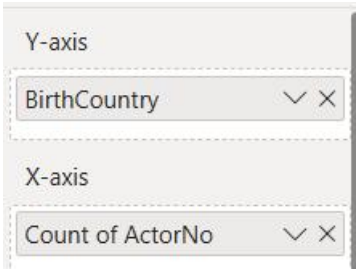




Figure 31: Visualization Pane for task Pass2l

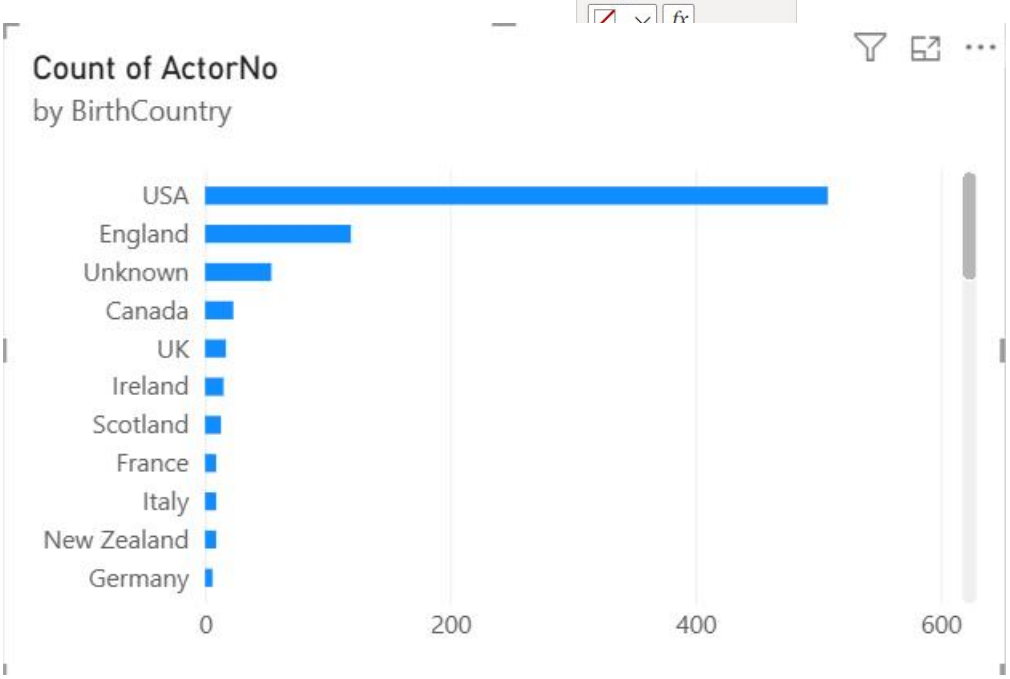


Figure 32: Clustered bar chart showing the count of actors by birth country.

M. Task Pass2m

In the final task of Pass part, I created new table named ***FavoriteSong5520,*** changed to ***Design View*** mode to fill in the three following fields including ***Field Name***, ***Data Type***, and ***Description (Optional).***

- ***Field Name***: SongID (Primary key), ***Data Type***: AutoNumber, ***Description***: Unique identifier for each song.
- ***Field Name***: SongTitle, ***Data Type***: Short Text, ***Description***: Title of the favorite song.
- ***Field Name***: Artist, ***Data Type***: Short Text, ***Description***: Name of the artist/band.
- ***Field Name***: ReleaseDate, ***Data Type***: Date/Time, ***Description***: Date when the song was released.

- ***Field Name***: StreamsMillion, ***Data Type***: Number, ***Description***: Total streams In million.

FavoriteSong5520

Field Name	Data Type	
SongID	AutoNumber	Unique identifier for each song
SongTitle	Short Text	Title of the favorite song
Artist	Short Text	Name of the artist/band
ReleaseDate	Date/Time	Date when the song was released
StreamsMillion	Number	al streams in million

Figure 33: Table Design View of FavoriteSong5520

Then, I entered 10 records for the FavoriteSong5520 Table.

FavoriteSong5520

SongID	SongTitle	Artist	ReleaseDate	StreamsMilli	Click to Add
1	Xoay Một Vòng	HIEUTHUHAI	8/15/2023	45	
2	Hẹn Gặp Em Dưới Ánh Trăng	HIEUTHUHAI	4/26/2022	82	
3	Sinh Nhật	HIEUTHUHAI	12/11/2023	34	
4	Người Im Lặng Gặp Người Hay Nói	HIEUTHUHAI	10/20/2022	56	
5	Không Thể Say (Live Action)	HIEUTHUHAI	5/1/2024	91	
6	Muộn Rồi Mà Sao Còn	Son Tung - MTP	4/29/2021	185	
7	Chúng Ta Của Tương Lai	Son Tung - MTP	8/3/2024	142	
8	Nơi Này Có Anh	Son Tung - MTP	2/14/2017	320	
9	Closer	The Chainsmokers	7/29/2016	3100	
10	Shape Of You	Ed Sheeran	6/1/2017	3850	

Figure 34: Date View of FavoriteSong5520 with 10 records

Next, in the tab ***Query***, choose ***Query Design,*** add ***FavoriteSong5520*** Table, drop down all columns in the table to ***Design Grid*** and enter ***[Enter Artist Name:]*** in

Field:	SongID	SongTitle	Artist	ReleaseDate	StreamsMillion
Table:	FavoriteSong5520	FavoriteSong5520	FavoriteSong5520	FavoriteSong5520	FavoriteSong5520
Sort:					
Show:	☑	☑	☑	☑	☑
Criteria:			[Enter Artist Name:]		
or:					

the criteria for the column ***Artist*** and saved query named ***FavoriteSongQuery5520.***

Figure 35: Query Design grid with parameter criteria

Subsequently, I entered value ***HIEUTHUHAI*** for the ***[Enter Artist Name:]***, the following figure showcase the result.

FavoriteSong5520 | FavoriteSongQuery5520

SongID	SongTitle	Artist	ReleaseDate	StreamsMilli
1	Xoay Một Vòng	HIEUTHUHAI	8/15/2023	45
2	Hẹn Gặp Em Dưới Ánh Trăng	HIEUTHUHAI	4/26/2022	82
3	Sinh Nhật	HIEUTHUHAI	12/11/2023	34
4	Người Im Lặng Gặp Người Hay Nói	HIEUTHUHAI	10/20/2022	56
5	Không Thể Say (Live Action)	HIEUTHUHAI	5/1/2024	91
(New)				0

Figure 36: Data View of the executed parameter query

Conclusion

After Successfully completed all Pass tasks from 2a to 2m, I have learn how to import and modify data, perform action queries, build parameter, select queries, integrate Access databases with Power BI to configure bidirectional cross-filtering relationships, generate various data visualizations (Table, Donut chart, Matrix, and Map), and design a custom Access table with associated records and queries.